Transformer Inference Profiler & KV-Cache / VRAM Inspector

Real-Time Telemetry, Entropy Dynamics, and Hardware Allocator Mechanics

- **Diagnostic Architecture:** Real-time interception of autoregressive decoding steps without GPU-to-CPU stalls.
- **Unified Observability:** Bridges information-theoretic predictive uncertainty with physical hardware memory dynamics.
- **Multi-Engine Integration:** Seamless profiling across PyTorch native CausalLM and local Ollama GGUF endpoints.
- **Interactive Dashboard:** Plotly Dash interface featuring Token Entropy Canvas and Candidate Competition View.

TELEMETRY METRICS

H(Pt) & KV

Shannon entropy, logit margins, analytical KV bytes, and CUDA VRAM fragmentation.

The Memory Wall in Autoregressive LLM Inference

Compute-Bound Pre-Fill vs. Memory-Bandwidth-Bound Token Generation

- **Operational Asymmetry:** Prompt pre-fill is compute-bound (GEMM), while sequential generation is strictly memory-bandwidth bound (GEMV).
- **Hardware Saturation:** Each generated token requires streaming the full parameter weights and expanding KV-cache activations through GPU SRAM.
- **Memory Bandwidth Equation:** Bandwidth demand scales as B_mem = (2*P + M_KV) / Latency.
- **Context Length Cliff:** Memory required for long-context KV activations rapidly eclipses static model weights.

THE 32k CLIFF

16.4 GB

Standard MHA on 7B model requires 16.4 GB solely for KV cache at 32k context.

Profiler Architecture & Telemetry Interception

Non-Blocking Step Hooks, In-Memory Storage, and Zero-Disk Telemetry

- **PyTorch Hooks:** Intercepts unnormalized logits via torch.inference_mode() with zero synchronization stalls.
- **Dynamic Ollama Client:** GGUF model introspection via /api/show and streaming latency telemetry via /api/generate.
- **In-RAM Zero-Disk Architecture:** /dev/shm tmpfs logs, circular RAM buffers, and in-memory SQLite buffer with zero physical disk wear.
- **Dynamic Token Scaling:** Unconstrained EOS generation and context scaling up to 131k tokens without sentence truncation.

DISK WRITES

0.00 MB/s

Pure in-RAM circular logging and tmpfs eliminate all physical drive wear.

KV-Cache Allocation: Analytical Math vs. PyTorch Reserved Pool

Attention Scaling Laws and Dynamic Caching Allocator Fragmentation

- **Exact Analytical Footprint:** M_KV = 2 * Batch * SeqLen * Layers * KV_Heads * HeadDim * DtypeBytes.
- **Grouped-Query Attention (GQA):** 8 KV heads vs 32 query heads achieves a 75% memory reduction over standard MHA.
- **Allocator Arena Dynamics:** PyTorch rounds allocations to 512B/2MB blocks and expands reserved pools in 20MB chunks.
- **Fragmentation Telemetry:** Tracks Phi_frag = 1 - (M_allocated / M_reserved) to identify out-of-memory risks before failure.

GQA SAVINGS

75%

Llama-3-8B reduces KV-cache from 16.4GB to 4.1GB at 32k tokens.

Token Entropy & Thermodynamic Softmax Scaling

Information-Theoretic Telemetry and Softmax Temperature Perturbation

- **Shannon Entropy:** Quantifies model predictive uncertainty: H(P_t) = -sum(p * log2(p)).
- **Thermodynamic Logit Inversion:** Reconstructs candidate logits dynamically via z_i = z_1 + T_eff * ln(p_i / p_1).
- **Probability Margins:** Delta_p = p(1) - p(2) reveals intense competition between alternative completion candidates.
- **Token Entropy Canvas:** Dual paragraph flow and pills view with cyan text selection and non-destructive inspection.

THERMODYNAMICS

T in [0.1, 2]

Simulates candidate probability redistribution across temperatures in real-time.

Latency Profiles & 60 FPS Visualizations

Pre-Fill vs Generation Latency with Adaptive Downsampling

- **Time-to-First-Token (TTFT):** Reflects prompt pre-fill latency across matrix-matrix attention kernels.
- **Inter-Token Latency (ITL):** Steady-state autoregressive decoding time with real-time jitter tracking.
- **Adaptive Downsampling:** Compresses dense series (>250 points) for 18.5x faster rendering (4.4ms vs 82ms).
- **Host RAM Safety:** Proactive memory estimation and 1-click emergency model eviction prevent laptop freezes.

CHART SPEEDUP

18.5x

Vectorized operations and downsampling deliver buttery-smooth 60 FPS rendering.

Real-World Architectural Case Studies

Empirical Telemetry across Llama-3, Qwen-2.5, and DeepSeek-V3

- **Llama-3-8B:** GQA (8 KV heads) exhibits linear KV growth reaching 1.02 GB at 8k context and 4.10 GB at 32k context.
- **Qwen-2.5-14B:** 48 layers with GQA requires 6.14 GB at 32k context, maintaining steady ITL of ~45 ms.
- **DeepSeek-V3 (MLA):** Multi-Head Latent Attention compresses KV representations into a 512-dim latent vector.
- **Compression Superiority:** MLA achieves an 86% memory reduction relative to standard MHA, scaling easily to 128k context.

MLA ADVANTAGE

2.2 GB

DeepSeek-V3 requires only 2.2 GB for 32k context vs 16.4 GB for MHA.

Roadmap & Future Extensions

Advanced PagedAttention Telemetry and Distributed Multi-GPU Serving

- **PagedAttention Integration:** Visualizing virtual block tables, page-level fragmentation, and memory sharing across parallel requests.
- **vLLM & SGLang Interceptors:** Direct hooks into production inference engine memory managers.
- **Multi-GPU Telemetry:** Profiling tensor-parallelism communication bubbles, NCCL collective overhead, and TP-rank memory skew.
- **Automated Speculative Decoding:** Tracking draft model acceptance rates and verification step latency savings.

NEXT FRONTIER

Distributed

Extending telemetry to multi-node Tensor Parallelism and Speculative Decoding.